Örnek 1

$$3-5.\left[2-\underbrace{(1-8)}_{-7}.4-3.(-5)\right]-2-5.\underbrace{(3-4)}_{-1}=?$$

$2+28+15$ $\qquad$ $-5.(-1)$

$-2+5=+3$

$3-5.45+3 \Rightarrow 3-225+3 \Rightarrow -219$

Örnek 2

$$(-3)^2-\left[\underbrace{(-30):3}_{-10}+\underbrace{2.(-5)}_{-10}-\underbrace{8.\overbrace{(2-5)}^{-3}}_{+24}\right].2-9=?$$

$9-(4).2-9=-8$

Örnek 3

$$x-\left\{1-\underbrace{[2x-3.(x+1)]}\right\}+4.(2-2x)=?$$

$$x-\left[1-(2x-3x-3)\right]+8-8x$$

$(-x-3)$

$x-(1+x+3) \Rightarrow x-1-x-3+8-8x = -4+8-8x \Rightarrow +4-8x$

Örnek 4

$$-3.\left[(x+2y)-3-2.(x+y)\right]-\left[(3x+1)-(x-8)\right]=?$$

$x+2y-3-2x-2y \quad -(3x+1-x+8)$

$2x+9$

$-3(-x-3)-2x-9$

$+3x+9-2x-9=x$

Örnek 5

$2\frac{1}{5} = ?$ $\quad \frac{2 \cdot 5 + 1}{5} = \frac{11}{5}$ //

Örnek 6

$\frac{5}{6} + \frac{4}{5} + \frac{2}{3} = ?$

(5) (6) (10)

$\frac{25}{30} + \frac{24}{30} + \frac{20}{30} = \frac{69}{30} = \frac{23}{10} = 2\frac{3}{10}$

Örnek 7

$5\frac{1}{2} - \frac{3}{5} = ?$

$\frac{5 \cdot 2 + 1}{2} - \frac{3}{5} = \frac{11}{2} - \frac{3}{5} = \frac{55}{10} - \frac{6}{10} = \frac{49}{10} = 4\frac{9}{10}$

(5) (2)

Örnek 8

1) $\frac{5}{8} \cdot \frac{3}{4} = ?$ $\quad \frac{15}{32}$ //

2) $\frac{2}{3} \cdot 5\frac{4}{5} = ?$

$\frac{2}{3} \cdot \frac{5 \cdot 5 + 4}{5} = \frac{2}{3} \cdot \frac{29}{5} = \frac{58}{15}$

3) $2 \cdot \frac{3}{5} = ?$

$\frac{2}{1} \cdot \frac{3}{5} = \frac{6}{5}$ //

Örnek 9

$2\frac{3}{4} \cdot \left(-3\frac{5}{6}\right) \cdot \frac{24}{11} = ?$

$\frac{11}{4} \cdot \left(-\frac{23}{6}\right) \cdot \frac{24}{11} = -23$ //

Örnek 10

$$\frac{3}{4} : \frac{2}{3} = ?$$

$$\frac{3}{4} \cdot \frac{3}{2} = \frac{9}{8} //$$

Örnek 11

$$2\frac{1}{5} : 1\frac{1}{3} = ?$$

$$\frac{11}{5} : \frac{4}{3} = \frac{11}{5} \cdot \frac{3}{4} = \frac{33}{20} //$$

Örnek 12

$$\left(2\frac{1}{3} + 1\frac{1}{4}\right) - \left(1\frac{1}{2} - 1\frac{1}{4}\right) = ?$$

$$\left(\frac{7}{3} + \frac{5}{4}\right) - \left(\frac{3}{2} - \frac{5}{4}\right)$$

(4) (3) (2)

$$\left(\frac{28}{12} + \frac{15}{12}\right) - \left(\frac{6}{4} - \frac{5}{4}\right) = \frac{43}{12} - \frac{1}{4} = \frac{43-3}{12} = \frac{40}{12} = \frac{10}{3} //$$

(3)

Örnek 13

$$\underbrace{\left[\left(-\frac{3}{4}\right) : \frac{2}{3}\right]} \cdot \underbrace{\left[\frac{5}{3} \cdot \left(-\frac{4}{3}\right)\right]} = ?$$

$$-\frac{3}{4} \cdot \frac{3}{2} = \frac{-9}{8} \cdot \frac{-20}{9} = \frac{20}{8} = \frac{5}{2} //$$

Örnek 14

$$\left(1\frac{2}{3}\right)^{-1} + \left(\frac{2}{7}\right)^{-1} = ?$$

$$\left(\frac{5}{3}\right)^{-1} + \left(\frac{2}{7}\right)^{-1}$$

$$\frac{3}{5} + \frac{7}{2} = \frac{6+35}{10} = \frac{41}{10}$$

(2) (5)

Örnek 15

$$\frac{\left(\frac{1}{5} : \frac{1}{3}\right) + \left(\frac{3}{4} - \frac{1}{8}\right)}{\frac{7}{8} : \frac{5}{7}} = ?$$

$$\frac{1}{5} \cdot \frac{3}{1} = \frac{3}{5}$$

$$\frac{3}{4} - \frac{1}{8} = \frac{6-1}{8} = \frac{5}{8}$$

(2)

$$\frac{7}{8} \cdot \frac{7}{5} = \frac{49}{40}$$

$$\frac{3}{5} + \frac{5}{8} = \frac{24+25}{40} = \frac{49}{40}$$

(8) (5)

$$\frac{\frac{49}{40}}{\frac{49}{40}} = \frac{49}{40} \cdot \frac{40}{49} = 1 //$$

Örnek 16

$$\left(\frac{1-1\frac{1}{3}}{1+1\frac{1}{3}}\right)^2 \cdot \left(\frac{7}{2}\right)^2 = ?$$

$$1-1\frac{1}{3} = 1-\frac{4}{3} = \frac{-1}{3}$$

$$1+1\frac{1}{3} = 1+\frac{4}{3} = \frac{7}{3}$$

$$= \left(\frac{\frac{-1}{3}}{\frac{7}{3}}\right)^2 \cdot \left(\frac{7}{2}\right)^2 =$$

$$\frac{-1}{\cancel{3}} \cdot \frac{\cancel{3}}{7} = \left(\frac{-1}{7}\right)^2 \cdot \left(\frac{7}{2}\right)^2 = \frac{1}{\cancel{49}} \cdot \frac{\cancel{49}}{4} = \frac{1}{4} //$$

Örnek 17

$a^{-1} = \frac{1}{5}$ ve $b^{-1} = \frac{1}{4}$ olduğuna göre $\underbrace{\frac{1}{a} - \frac{1}{b}}$ ifadesi neye eşittir?

$$\frac{1}{a} = \frac{1}{5} \qquad \frac{1}{b} = \frac{1}{4}$$

$$\frac{1}{5} - \frac{1}{4} = \frac{4-5}{20} = -\frac{1}{20} //$$

Örnek 18

$$1+\cfrac{1}{1+\cfrac{1}{1+\frac{1}{3}}} = ?$$

$$1+\frac{1}{3} = \frac{4}{3}$$

$$1+\cfrac{1}{\frac{4}{3}} = 1+\frac{3}{4} = \frac{7}{4}$$

$$1+\cfrac{1}{\frac{7}{4}} \rightarrow \frac{4}{7} = 1+\frac{4}{7} = \frac{11}{7} //$$

Örnek 19

$$1+\cfrac{\frac{1}{2}+\frac{2}{3}}{1+\cfrac{1}{1+\frac{1}{2}}} = ?$$

$$\frac{1}{2}+\frac{2}{3} = \frac{1}{2}+\frac{2}{1}\cdot\frac{1}{3} = \frac{1}{2}+\frac{8}{3} = \frac{3+16}{6} = \frac{19}{6}$$

$$1+\cfrac{1}{1+\frac{1}{2}} = 1+\frac{1}{1}\cdot\frac{2}{3} = 1+\frac{2}{3} = \frac{5}{3}$$

$$1+\cfrac{\frac{19}{6}}{\frac{5}{3}} = 1+\frac{19}{6}\cdot\frac{3}{5} = 1+\frac{57}{30} = \frac{87}{30} = \frac{29}{10} //$$

Örnek 20

Aşağıdaki kesirli sayıları virgül kullanarak gösteriniz.

$\frac{2}{10} = 0,2$ $\quad \frac{3}{100} = 0,03$ $\quad 4\frac{75}{100} = \frac{475}{100} = 4,75$

$\frac{125}{10} = 12,5$ $\quad \frac{375}{1000} = 0,375$

Örnek 21

Aşağıda verilen ondalık sayıları kesirli sayı olarak yazınız.

$2,4 = \frac{24}{10} = 2\frac{4}{10}$ $\quad 0,0001 = \frac{1}{10000}$

$5,125 = \frac{5125}{1000} = 5\frac{125}{1000}$

Örnek 22

$\frac{2}{5}$ kesrini ondalık kesir olarak yazınız.

$\frac{2}{5} = \frac{4}{10} = 0,4$

$\begin{matrix}5\\(2)\end{matrix}$

Örnek 23

$\frac{1}{4}$ kesrini ondalık kesir olarak yazınız.

$\frac{1}{4} = \frac{25}{100} = 0,25$

$\begin{matrix}4\\(25)\end{matrix}$

Örnek 24

$12,37 + 2,084 = ?$

$$\begin{array}{r} 12,37 \\ +\ 2,084 \\ \hline 14,454 \end{array}$$

Örnek 25

$8,04 - 3,325 = ?$

$$\begin{array}{r} {\scriptstyle 10\ \ 3\ \ 10} \\ 7\not{8},\not{0}\not{4}\not{0} \\ 3,325 \\ \hline 4,715 \end{array}$$

Örnek 26

$36,5 \times 0,24 = ?$

$$\begin{array}{r} 36,5 \\ \times\ 0,24 \\ \hline 1460 \\ 730 \\ 000 \\ \hline 08,760 \end{array} \quad \begin{array}{l} \text{2 x 4} \\ \text{1} \end{array}$$

$= 8,76$ //

Örnek 27

$$\frac{0,02}{0,004}_{(1000)} + \frac{0,03}{0,005}_{(1000)} + \frac{0,04}{0,08}_{(100)} = ?$$

$$\frac{20}{4}_{(10)} + \frac{30}{5}_{(8)} + \frac{400}{8}_{(5)} =$$

$$\frac{200}{40} + \frac{240}{40} + \frac{2000}{40} = \frac{2440}{40} = \frac{122}{2}$$

$= 61$ //